

DATA ANALYTICS INTERNSHIP

Assignment 1: Excel Functions, Pivot Tables & SQL Queries

Setup Requirement

Set up the following before starting:

- Microsoft Excel (2016 or later) or Google Sheets / LibreOffice Calc as alternative.
- **MySQL Workbench** or **SQLite DB Browser** or any online SQL editor (e.g. db-fiddle.com, sqLiteonline.com).

Task 1: Excel Formatting, Functions & Charts

Dataset to be Entered in Excel (Save as sales_data.xlsx)

Open a new Excel file and enter the following data starting from cell A1. Add column headers in the first row.

emp_id	name	department	product	quantity	unit_price	sale_date
E01	Aarav	Electronics	Laptop	2	55000	2025-02-03
E02	Bhavya	Fashion	Saree	5	4500	2025-02-05
E03	Chirag	Electronics	Headphones	10	1500	2025-02-06
E04	Diya	Fashion	Shoes	3	2200	2025-02-07
E05	Eshan	Electronics	Smartphone	4	32000	2025-02-09
E01	Aarav	Electronics	Tablet	1	18000	2025-02-11
E06	Farah	Grocery	Rice (kg)	20	80	2025-02-12
E07	Gaurav	Fashion	Watch	2	3500	2025-02-14
E08	Hina	Grocery	Oil (litre)	15	180	2025-02-15
E03	Chirag	Electronics	Mouse	8	899	2025-02-18
E02	Bhavya	Fashion	T-shirt	10	499	2025-02-20
E09	Ishan	Grocery	Sugar (kg)	25	55	2025-02-22
E04	Diya	Fashion	Backpack	4	1800	2025-02-24
E05	Eshan	Electronics	Charger	12	650	2025-02-26
E10	Jiya	Grocery	Tea (gm)	5	350	2025-02-28

1.1 Cell Formatting

Apply the following formatting and submit a screenshot:

- Make headers **Bold**, centred, with a light blue background.
- Apply **borders** to all cells of the data range.
- Format **unit_price** column as Currency (INR, no decimals).
- Format **sale_date** column as Date in **DD-MMM-YYYY** format.

1.2 Add a New Column: total_amount

Add a new column **H** named **total_amount**. In cell H2, type the formula: **=E2*F2** (quantity x unit_price). Drag it down for all rows.

1.3 Useful Excel Functions

In separate cells below the data, calculate and label each of the following:

#	Function	Excel Formula to Use	Label
1	Total revenue	=SUM(H2:H16)	Total Revenue:
2	Average sale value	=AVERAGE(H2:H16)	Average Sale:
3	Max sale value	=MAX(H2:H16)	Maximum Sale:
4	Min sale value	=MIN(H2:H16)	Minimum Sale:
5	Count of Fashion sales	=COUNTIF(C2:C16,"Fashion")	Fashion Sales Count:
6	Sum of Electronics revenue	=SUMIF(C2:C16,"Electronics",H2:H16)	Electronics Revenue:
7	Bonus column (IF total > 50000 then High > 50000, else Normal)	=IF(H2>50000,"High","Normal")	Add as column I

1.4 Charts and Graphs

Create the following 3 charts and submit clear screenshots:

- **Bar Chart:** Product Names vs Total Amount
- **Pie Chart:** Total Revenue Share by Department
- **Line Chart:** Sale Date vs Total Amount (sorted by date)

Each chart must have a **title, axis labels, and a legend** where applicable.

Task 2: Pivot Tables & Pivot Charts in Excel

Continue using the same dataset from Task 1.

2.1 Create Pivot Tables

Insert > PivotTable. Create the following 3 pivot tables and submit screenshots:

#	Pivot Table	Rows	Columns	Values
1	Revenue by Department	department	(none)	Sum of total_amount
2	Revenue by Department and Employee	department, name	(none)	Sum of total_amount
3	Quantity sold by Product	product	department	Sum of quantity

2.2 Pivot Chart

Using Pivot Table #1, create a **Pivot Chart** (Bar or Column). Add a title and submit a screenshot.

2.3 Apply a Slicer

Add a **Slicer** to your Pivot Table for the **department** field. Use the slicer to filter results and take a screenshot showing the slicer in use.

Task 3: SQL Queries (Practice Database)

Use any SQL editor (MySQL Workbench, SQLite DB Browser, or online tools like db-fiddle.com or sqliteonline.com). Create two tables using the schema and INSERT statements below. Then write and run the queries listed.

Schema and Sample Data (Run these CREATE / INSERT statements first)

```
CREATE TABLE employees (
  emp_id INT PRIMARY KEY,
  name VARCHAR(50),
  department VARCHAR(30),
  salary INT,
```

```

        hire_date DATE
    );

INSERT INTO employees VALUES
(1, 'Aarav', 'Electronics', 55000, '2022-01-10'),
(2, 'Bhavya', 'Fashion', 48000, '2021-05-15'),
(3, 'Chirag', 'Electronics', 52000, '2023-03-20'),
(4, 'Diya', 'Fashion', 72000, '2020-07-01'),
(5, 'Eshan', 'Electronics', 60000, '2022-11-12'),
(6, 'Farah', 'Grocery', 45000, '2023-08-05'),
(7, 'Gaurav', 'Fashion', 85000, '2019-04-22'),
(8, 'Hina', 'Grocery', 42000, '2024-01-18'),
(9, 'Ishan', 'Grocery', 55000, '2022-09-10'),
(10, 'Jiya', 'Electronics', 78000, '2020-12-01');

CREATE TABLE sales (
    sale_id INT PRIMARY KEY,
    emp_id INT,
    product VARCHAR(50),
    amount INT,
    sale_date DATE
);

INSERT INTO sales VALUES
(101, 1, 'Laptop', 110000, '2025-02-03'),
(102, 2, 'Saree', 22500, '2025-02-05'),
(103, 3, 'Headphones', 15000, '2025-02-06'),
(104, 4, 'Shoes', 6600, '2025-02-07'),
(105, 5, 'Smartphone', 128000, '2025-02-09'),
(106, 1, 'Tablet', 18000, '2025-02-11'),
(107, 6, 'Rice', 1600, '2025-02-12'),
(108, 7, 'Watch', 7000, '2025-02-14'),
(109, 8, 'Oil', 2700, '2025-02-15'),
(110, 3, 'Mouse', 7192, '2025-02-18');

```

3.1 Write SQL Queries

Write and execute the following queries. Submit the query and a screenshot of its output:

#	Query Description
1	Display all rows of the employees table (basic SELECT).
2	List employees from the 'Fashion' department only (WHERE clause).
3	Display all employees sorted by salary in descending order (ORDER BY).
4	Display the names of employees whose salary is between 50000 and 70000 (BETWEEN).
5	Find the total number of employees in each department (GROUP BY + COUNT).
6	Find the average salary per department (GROUP BY + AVG).
7	Find departments where the average salary is greater than 50000 (GROUP BY + HAVING).
8	JOIN: Show emp_id, name, product, and amount for all sales (INNER JOIN sales and employees).
9	Find the total sales amount made by each employee (JOIN + GROUP BY + SUM).
10	Subquery: Find employees whose salary is greater than the average salary of all employees.

3.2 CRUD Operations (Create, Update, Delete)

Write and execute the following statements. Verify with a SELECT after each:

#	Operation
1	INSERT: Add a new employee (emp_id=11, name='Kavya', department='Fashion', salary=50000, hire_date='2025-03-01').

2	UPDATE: Increase the salary of all 'Grocery' department employees by 10%.
3	DELETE: Remove the employee with emp_id=11 (the one inserted in step 1).

Deliverables

- Task 1: Excel file (or screenshots) showing the formatted dataset, total_amount column, 7 calculated function values, and the 3 charts.
- Task 2: Screenshots of 3 Pivot Tables, 1 Pivot Chart, and the Slicer in use.
- Task 3: All 10 SQL queries plus 3 CRUD statements with their outputs (screenshots).

How to Submit

1. Create the assignment in MS Word (.doc / .docx) or PDF format.
2. Include your full name, course / internship name, and assignment number at the top of the document.
3. Paste your formulas, SQL queries, and clear screenshots of every output for each task.
4. Upload the completed file (and your Excel file) to your Google Drive.
5. Enable sharing access by setting the file to "Anyone with the link can view".
6. Copy the Google Drive link and submit it before the assignment deadline through the official submission portal.